

Session 111: Biomedical and Neural Signal Learning



SPCOM 2026

International Conference on
Signal Processing and Communications

TIME 9:30 AM – 11:00 AM IST VENUE SP 1.20 (TVS Lab)

111.1

Uncertainty-Aware Structural Attention Features for Hallucination Detection in LLMs

Rajahmundry Lokesh, Pagadala Usha

111.2

WASSL-EGR: Wavelet Attention and State-Space Learning for Robust EMG-based Gesture Recognition

Mainak Ghosh, Anup Nandy

111.3

ProactiveGuard: IoT-Driven Fall Detection and RAG-Based Caregiver Assistance for Alzheimer's Patients

Naman Omar, Vengadeswaran Shanmugasundaram, Mohamedsipli M

111.4

IF-EWT-based framework for classification of motor imagery EEG signals

Ram Bilas Pachori, Krishnakant Sharma

111.5

Unified Adaptive Quantization and Representation-Aware Caching for Diffusion Models

Deepanshi Sharma, Rudra Pratap, Shivam Jaiswal, Parimala Kancharla

111.6

Shared Neural Dynamics Between Overt and Inner Speech: An EEG Signal Analysis

Aastha Kachhi

111.7

Decoding Silent Reading: A Novel EEG-to-Text Method

Noah Thomas, K J Theophene Lynn, Divya James, Jyoti Nigam

111.8

EEG-based Functional Connectivity During Sleep As A Biomarker for Disorders of Consciousness

Anusha A S

111.9

EventF2S: Asynchronous and Sparse Spiking AER Framework using Neuromorphic-Friendly Algorithm

Lakshmi Annamalai, Chetan Singh Thakur

111.10

Learning Neuromuscular Phase Transitions in Yoga Postures from Multi-Muscle EMG Signals Using Deep Temporal Latent Representations

Suvitti, Mohan Raghavan

111.11

fNIRS contains usable stimulus information much earlier than the canonical HRF peak

Suranjita Ganguly, Malaika Chhaya

111.12

ECG-CaRE: An Interpretable and Edge-Deployable Framework for Arrhythmia Classification Competitive with Deep Models

Nilanjan Biswas, Suman Samui

111.13

Spoken Response-Based Automatic Screening of Suicidal Risk Among Adolescents

Rohan Kumar Gupta

111.14

Multimodal Signal Analysis of Simulated Microgravity on Cerebral-Systemic Hemodynamics and Cardiac Autonomic Regulation

Debanjan Das

111.15

Multimodal Speaker Separation in Audio-Visual Scene using Spherical Microphone Array

Jayesh Upadhyay, Priyadarshini Dwivedi, Gyanajyoti Routray, Rajesh M Hegde

111.16

Breaking the Pair: Evaluating Dyadic Interaction via Speaker Switching

Nishchay Nilabh, Neeraj Sharma

Session 112: Speech and Spoken Language Processing

TIME 9:30 AM – 11:00 AM IST VENUE SP Building Side Foyer



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- 112.1** **Voice Conversion-Driven Speaker Anti-Spoofing Framework Using Attention-Based Seq2Seq Modeling**
Govind Khandelwal, Rantu Buragohain, Karan Nathwani
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- 112.2** **Pronunciation based Intelligibility Assessment of Dysarthric Speech**
Komal Bharti, Pradip Das, Hemant Patil
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- 112.3** **A Comparative Study of Pre-trained Speech Encoders and Training Objectives for Large-Scale Indic Spoken Language Identification**
Agneeth Basu, Pavan Kumar J, Sujith Pulikodan, Prasanta Ghosh
-
- 112.4** **Adapting Whisper Models using LoHA for Robust Recognition of Children's Speech**
Ankita, Syed Shah Nawazuddin
-
- 112.5** **S-Transform Features for Automatic Classification of Phonation Types**
Doondi Bala Saraswathi Aluru, Gurugubelli Krishna
-
- 112.6** **RG-RCT: Real-Governed Representation-Calibrated Training for Low-Resource Medical ASR**
Sri Charan Devarakonda, Ravi Sastry Kolluru, Srihari Bandarupalli, Bhavana Akkiraju, Anil Vuppala
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- 112.7** **ResGDS-HLF: Residual Gumbel Dimension Selection with Hierarchical Layer Fusion for Stuttering Event Classification**
Pragya Khanna, Vijaya Saraswathi R, Vasavi Ravuri, Anil Vuppala
-
- 112.8** **Architectural Control of Phonetic Invariance via Additive U-Net Multi-Task Learning**
Vikram Lakkavalli, Neelam Sinha
-
- 112.9** **MRAN-UNet: Physics-Informed Harmonic Frequency Attention for Multilingual Speech Enhancement**
Satya Prasad G, Puja Bharati, Sabyasachi Chandra, Shyamal Das Mandal, Nirmalya Ghosh, Satya Prasad Gaddamedi
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- 112.10** **Are Speech Language Models More Noise Robust for Zero Shot Children's Speech Recognition**
Arindam Das, Abhijit Sinha, Hemant Kathania
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- 112.11** **Speech-to-Speech Translation Evaluation: A Systematic Analysis and Multi-Dimensional Metrics**
Lalaram Arya, Mrinmoy Bhattacharjee, Mahadeva Prasanna
-
- 112.12** **Neuro-Symbolic ASR for Dysarthric Speech via Articulatory-Constrained Posterior Reshaping**
Nishid S R, Jyothish Lal G

Session 113: Wireless Network Performance Analysis

TIME 9:30 AM – 11:00 AM IST VENUE ECE 1.07



113.1

SINR Analysis of an IRS-Assisted MU-MISO System

Lakshmi Jayalal, Shashank Shekhar, Sheetal Kalyani

113.2

Caching Performance under Constraints on Cache Updates

Chinmaya Venkatesh, Avhishek Chatterjee

113.3

Version Age in Multi-Process Gossip Networks under Distributed Sensing

Anuj Meena, Amitalok Budkuley

113.4

Statistical Fidelity, Computational Cost, and Moment-Matched Surrogates for FSO Turbulence Channels

Akhil Misra, Sainath Bitragunta

113.5

On minimization of average AoI for static age-threshold irregular repetition slotted ALOHA

Minha Mubarak, Vineeth Bala Sukumaran

113.6

Enhancing FSO Links Using Multi-HAPS and Opportunistic Relay Selection over IGGG Channels

Aishanee Chatterjee, Prashant Sharma, Swaminathan Ramabadrani

113.7

Optimizing Least Squares Channel Estimation using Deep Learning in 5G NR PUSCH

Peddinti T N Mrunalini, Moyella Saikumarreddy, Pradeep Goutam, Naru Pradeep

113.8

Latency-Control Tradeoffs in Communication-Control Co-Designed Networks

Songita Das, Gourab Ghatak, Geethu Joseph

113.9

Mobility Aware CSI Reliability in PASS Assisted Highway V2I Systems

Ura Modi, Dhaval Patel

113.10

Delay and Throughput Analysis of Prioritized EDCA in Latency-Sensitive Wi-Fi 8 Networks

Sreelakshmi Manjunath